

AI-Powered Credit Risk Intelligence Platform

An end-to-end credit-risk solution on the Home Credit Default Risk dataset — calibrated ML scoring, SHAP explainability, model-derived business rules, and an LLM natural-language → SQL interface, delivered as one interactive application.

307,511

APPLICANTS

0.763

ROC-AUC

5

INTEGRATED MODULES

1

DOCKER COMMAND

Credit decisions must be accurate and explainable

Lenders need default-risk assessment that survives credit-committee review, internal audit, and regulatory scrutiny. A single black-box score does not — the decision has to be traceable and usable by non-technical staff.

THE GAP

- Risk teams need a defensible number and the reasons behind it
- Analysts need readable policy, not model internals
- Business users need to interrogate data without writing SQL

WHAT THE PLATFORM DELIVERS

- Calibrated default probability + Low / Medium / High band
- Per-applicant SHAP reason codes
- Human-readable risk rules for analyst review
- Plain-English querying of the applicant data
- One Streamlit app, reproducible via Docker Compose

HOW IT FITS TOGETHER

System architecture

Home Credit dataset · 307K applications · loaded into DuckDB

ML PIPELINE

Preprocessing & feature engineering

↓

LightGBM champion + Logistic Regression baseline

↓

Isotonic probability calibration

↓

Risk bands · SHAP explanations · surrogate rules

TALK-TO-DATA PIPELINE

Schema-grounded prompt context

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Gemini NL → SQL generation

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Read-only SQL validator (allow-list, row cap)

↓

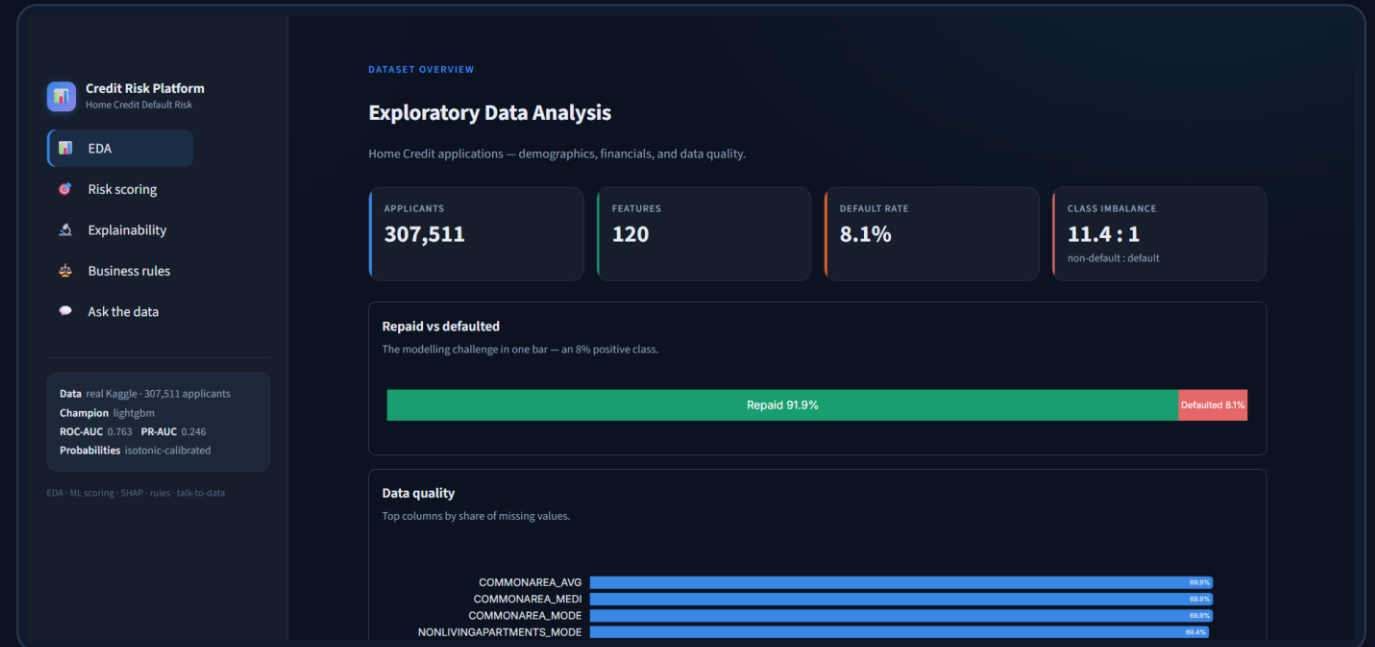
DuckDB query → business-readable answer

Streamlit UI · packaged with Docker Compose

EXPLORATORY DATA ANALYSIS

307K applications, ~8% default — a hard class imbalance

- **Imbalance.** ~8% positives — ROC-AUC / PR-AUC, not accuracy, drive evaluation
- **Missingness.** up to ~70% in housing columns; kept as signal, not imputed away
- **Anomaly.** ~18% of rows carry a 'not employed' placeholder → explicit flag
- **Every finding** is an exploratory association, not a causal claim



External credit scores are the strongest default signal

— **EXT_SOURCE.** 19.9% → 2.6% default across score deciles

— **Age.** risk falls steadily; 18–25 highest

— **Education.** 10.9% lower-secondary vs 1.8% academic degree

— **Family status** and car ownership add smaller, consistent gaps



LightGBM champion, calibrated to a real probability

MODEL	ROC-AUC	PR-AUC
Logistic Regression (baseline)	0.750	0.235
LightGBM + isotonic calibration	0.763	0.246

- **Class imbalance.** at ~8% positives, accuracy misleads — ranking metrics select the model
- **Calibration.** isotonic regression anchors scores to the true base rate, so a 0.30 reads as ~30%
- **Bands.** off the calibrated probability — Low < 0.20 · Medium $0.20\text{--}0.50$ · High ≥ 0.50
- **Version-proof scoring.** calibrator unwrapped and re-applied explicitly at inference

Validation fold: 61,503 applicants · default prevalence 8.07%

RISK SCORING

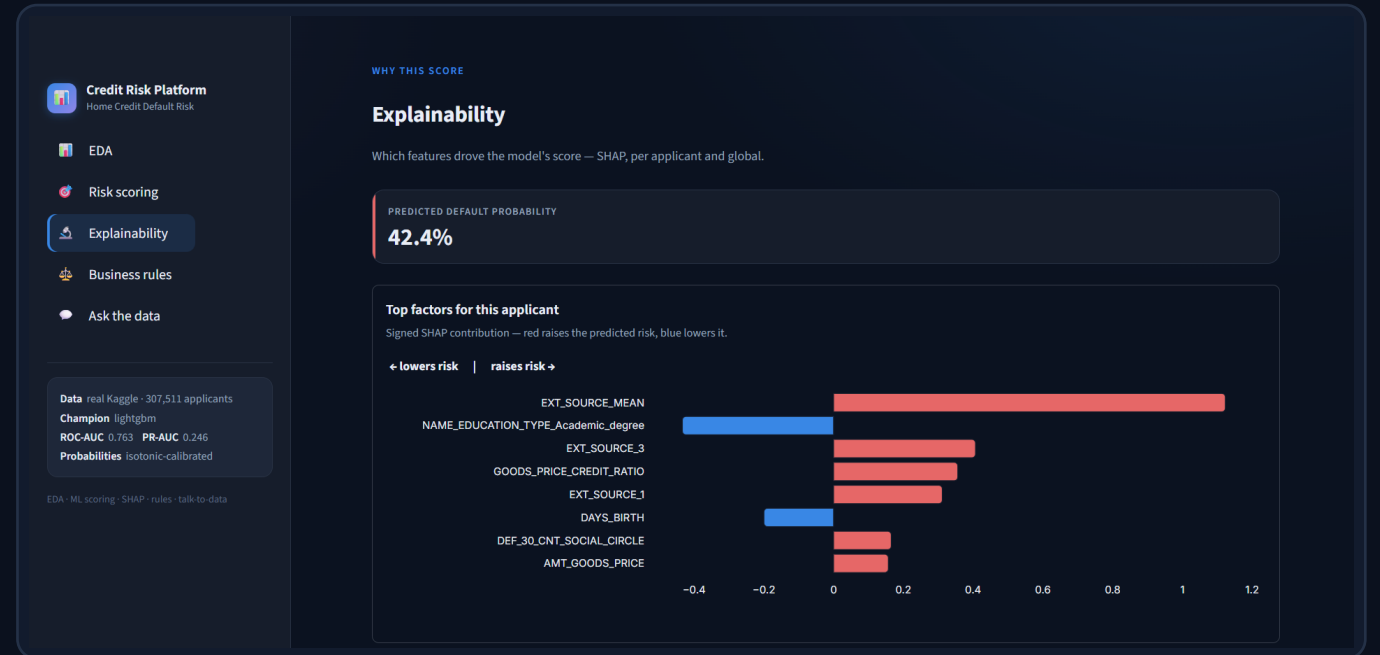
A defensible number for every applicant

- **Two entry modes.** sample a real applicant or enter one manually
- **Calibrated gauge.** probability shown as a real rate
- **Risk band.** Low / Medium / High with the driving attributes
- **Hand-off.** the scored applicant flows into Explainability and Rules



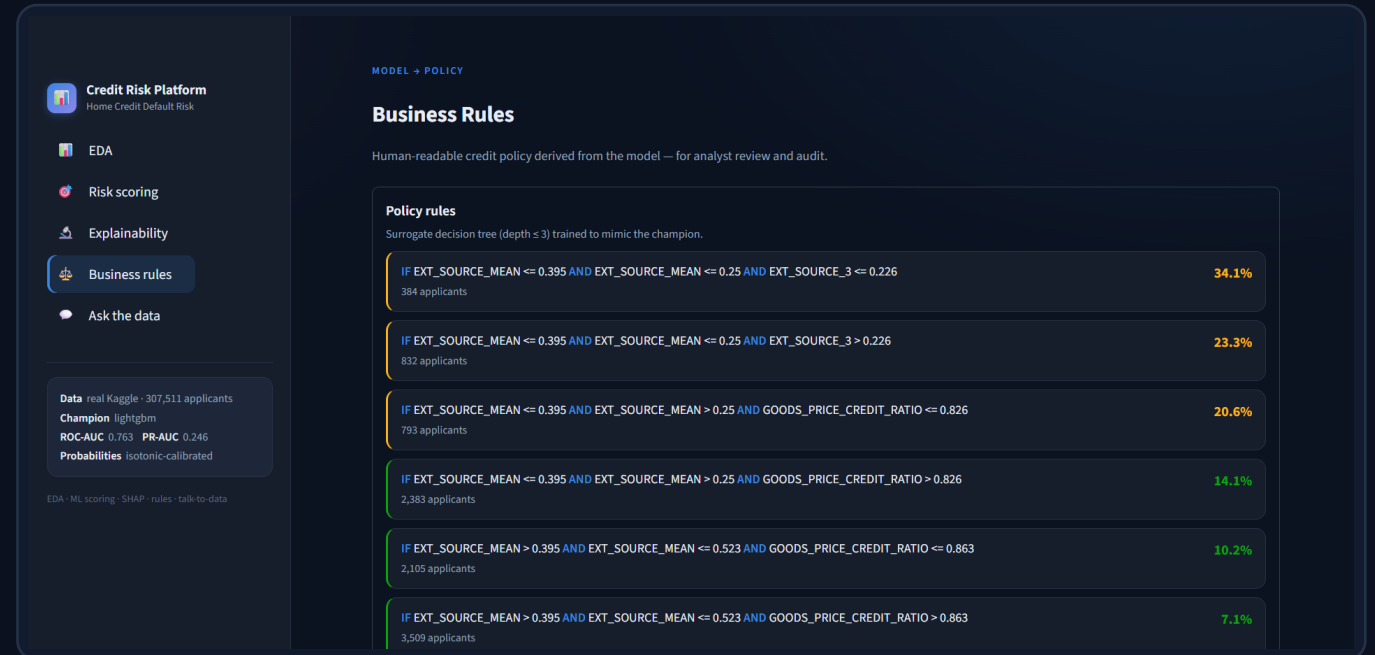
Every score ships with its reasons

- **SHAP TreeExplainer** on the LightGBM model — exact, fast
- **Local view.** signed per-applicant contributions; red raises risk, blue lowers it
- **Global view.** mean-|SHAP| over a 1,000-applicant sample
- **EXT_SOURCE features** dominate both local and global rankings



A human-readable approximation of the model

- **Surrogate tree.** depth ≤ 3 , trained to mimic the champion
- **Each leaf** is an IF / AND rule with its observed default rate (2.4% → 34.1%)
- **Companion table.** observed default rate per EXT_SOURCE band vs real labels
- **Decision support,** explicitly not lending policy



Ask the applicant data in plain English

- **Schema-grounded prompt** → Gemini generates SQL
- **Read-only validator.** DROP / DELETE / UPDATE / INSERT / ALTER rejected; approved tables only; row cap
- **Pre-verified templates** answer common questions with no LLM call
- **Transparency.** generated SQL + result table always shown

Credit Risk Platform
Home Credit Default Risk

- EDA
- Risk scoring
- Explainability
- Business rules
- Ask the data**

Data real Kaggle · 307,511 applicants
Champion lightgbm
ROC-AUC 0.763 **PR-AUC** 0.246
Probabilities isotonic-calibrated

EDA · ML scoring · SHAP · rules · talk-to-data

Sample questions
One click — pre-verified SQL, no LLM call.

- What is the overall default rate?
- What is the average income of applicants who defaulted vs did not?
- Show default rate by education level
- How many applicants own both a car and a house?
- What are the top 5 organization types by number of applicants?

What is the overall default rate?

Returned 1 row(s) across columns: default_rate_pct, n_applicants.

SQL used

```
SELECT AVG(TARGET) * 100 AS default_rate_pct, COUNT(*) AS n_applicants FROM applications
```

default_rate_pct	n_applicants
8.0729	307511

Ask about the applicant data...

Limitations & what's next

CURRENT LIMITATIONS

- Application table only — no bureau / previous-loan aggregates
- Calibration uses the evaluation fold
- Hyperparameter search intentionally limited
- Free-form NL → SQL needs an external API key
- Rules approximate the model — not formal policy
- No fairness audit or production banking validation

NEXT STEPS

- Aggregate bureau & previous-application history into features
- Automated hyperparameter optimization
- Dedicated train / calibration / test splits
- Model-drift monitoring
- Fairness & bias evaluation
- API inference service with authentication

SUMMARY

Built with a lean, reproducible stack

Data processing	Python · Pandas · NumPy
Analytical database	DuckDB
Machine learning	scikit-learn · LightGBM
Explainability	SHAP
LLM · SQL safety	Google Gemini · sqlparse
UI · visualization	Streamlit · Plotly
Deployment	Docker · Docker Compose · Streamlit Community Cloud

IN ONE LINE

Data → EDA → feature engineering →
calibrated ML → explainability →
business rules → NL-to-SQL → UI →
Docker.

The full AI-engineering lifecycle in one app.

LIVE DEPLOYMENT

Try the platform

<https://ai-credit-risk-intelligence-platform.streamlit.app/>

Deployed on Streamlit Community Cloud · committed champion model · schema-accurate synthetic fallback dataset

Source code

github.com/jinishar/credit_risk